DAVE CARROLL

PLATFORM ENGINEER

ABOUT

Platform engineer who builds automation for process improvement and a clearer understanding of systems. Three decades in IT, from Windows infrastructure and identity to DevOps, with a long record of sharing knowledge through clear documentation, talks, and blogging.

EXPERIENCE

Undisclosed Mortgage
Aug 2020 – Present
California

Senior DevOps Engineer / Senior Platform Engineer
Oct 2024 – Present

- A subject-matter expert for Windows Server, Active Directory, and Microsoft SQL Server.
- Builds Windows Server AMIs with HashiCorp Packer and PowerShell provisioning scripts, including updated build pipelines.
- Manages multiple Chef organizations, each with unique cookbooks, environments, and roles.
- Redesigned Chef cookbooks to be organization-agnostic, with environment-specific details applied through Chef environment files and roles, and added performance improvements such as in-memory caching during client runs; developed Cursor rules for the team's cookbook development.
- Uses Gemini and Cursor daily across development, troubleshooting, and automation work.
- Provisions AWS resources with OpenTofu.
- Mentors junior engineers and interns on DevOps practices, automation scripting, and AWS resource management.

DevOps Engineer / Senior DevOps Engineer
Aug 2020 – Oct 2024

- Initially worked without a general team; joined a vertical team supporting one division in April 2021.
- Supported dozens of in-house web applications for that division, on AWS with CloudFormation, Troposphere, Chef, and PowerShell.
- Led RDS cost reduction and right-sizing efforts across the division's AWS environment.
- Established the migration pattern from Windows Server 2012 R2 (Chef 12) to Windows Server 2022 (Chef 17) and completed the majority of the migrations.
- Wrote PowerShell modules to simplify working with AWS Auto Scaling Groups, other AWS services, and Chef.
- Joined an on-premises-to-cloud migration effort in its later stages, supporting ongoing moves to AWS.

CONTACT

Greater Nashville Area, Tennessee, US

WEBSITE
thedavecarroll.com

GITHUB
thedavecarroll

PROJECTS

BluebirdPS (Open-Source PowerShell Module)
May 2020 – Dec 2024

BluebirdPS is an open-source PowerShell 7 module that provides an automation client for the Twitter API. The repository was archived in December 2024.

PoShEvents (Open-Source PowerShell Module)
May 2018 – Present

PoShEvents is an open-source PowerShell module which assists users with querying the Windows event logs. It also provides a function to write structured data to event entries.

PoShGroupPolicy (Open-Source PowerShell Module)
Mar 2018 – Present

PoShGroupPolicy is an open-source PowerShell module that deciphers several group policy settings into easy to save, search, and manipulate custom PowerShell objects.

PUBLICATIONS

The PowerShell Conference Book Volume 3
Leanpub · Aug 2020

#PS7Now
Leanpub · Mar 2020

The PowerShell Conference Book Volume 2
Leanpub · Jan 2020

PRESENTATIONS

University Nov 2019 – May 2020
California

IAM Applications Developer

- Built a C# Web API for managing Active Directory groups, membership and user attributes.
- Automated its builds in Azure DevOps and its releases across environments with Octopus Deploy.
- Improved ADFS logon-failure logging in PowerShell.

Undisclosed Global Financial Oct 2018 – Oct 2019
Greater Nashville Area, TN

Identity Management and Automation Engineer

- Collaborated with a team spanning the US, UK, Switzerland, Germany, China, and Singapore, writing PowerShell automation for build configuration and validation checks.
- Deployed Active Directory infrastructure changes under formal change management, including new domain controllers and GPO updates via Microsoft's Advanced Group Policy Management (AGPM).

Undisclosed Medical Supply Sep 2016 – Oct 2018
Greater Nashville Area, TN

Systems Engineer, Infrastructure

- Built a PowerShell migration workflow that moved roughly 2,000 mailboxes from two on-premises domains to Exchange Online, preparing AD objects, licensing target accounts, and copying mailbox contents; used for the initial cutover and later ad-hoc migrations.
- Completed an AD modernization project: redesigned AD Sites for a smaller forest, migrated SYSVOL replication from FRS to DFS-R, extended the schema with custom attributes, and wrote a user synchronization script to provision mirrored accounts in a resource/trust domain.
- Established group policy standards, wrote startup scripts to deploy required applications, and wrote PowerShell scripts for reporting, environment documentation, and data integrity checks.

University Jun 2012 – Apr 2016
California

Undisclosed Community College Nov 1998 – May 2012
Tennessee

Engineering Staffing Company Nov 1997 – Sep 1998
Jackson, Tennessee

C&M Computer Center Nov 1994 – Nov 1997
Jackson, Tennessee

Writing Cross-Platform PowerShell
SQL Saturday Atlanta 2025 - AI & BI · Mar 2025

Writing Cross-Platform PowerShell
The PowerShell + DevOps Global Summit 2024 · Apr 2024

BluebirdPS: From Nesting to Hatched and Flying
The PowerShell + DevOps Global Summit 2023 · Apr 2023

Your Code is Flawless, But How YOU doing?
The PowerShell + DevOps Global Summit 2023 · Apr 2023

Creating BluebirdPS: A PowerShell 7 Twitter Automation Client
The PowerShell + DevOps Global Summit 2021 · Apr 2021

Crossing the Divide: A System Administrator's Path to DevOps
Research Triangle PowerShell User Group - PowerShell Saturday · Sep 2019

EDUCATION

University of Illinois Springfield
English Language and Literature, General · 2006 – 2007

Jackson State Community College
A.S., Information Systems · 2000 – 2002

University of Memphis
English Language and Literature, General · 1988 – 1990

CERTIFICATES

MCPS: Microsoft Certified Professional (Windows Server 2012)
Microsoft · Feb 2014

MCSE1: Windows NT 4.0
Microsoft · Dec 2000

CompTIA Network+ Certification
CompTIA · Nov 2000

CompTIA A+ Certification
CompTIA · Feb 1996